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解: 1) $G(s) = \frac{K}{(T_1s+1)(T_2s+1)(T_3s+1)}$

$G(s)$ 正实部极点数 $P=0$

$N_+ = 0 \quad N_- = 1 \quad R = 2(N_+ - N_-) = -2$

$Z = P - R = 2$ 系统闭环不稳定, 有两个正实部极点数

2) $G(s) = \frac{K}{s(T_1s+1)(T_2s+1)}$

$G(s)$ 正实部极点数 $P=0$

$N_+ = 0 \quad N_- = 0 \quad R = 2(N_+ - N_-) = 0$

$Z = P - R = 0$ 系统闭环稳定

$V=1$, 从奈氏曲线上 $W=0^+$ 的对应点起逆时针补作 90° 且半径为无穷大的虚圆弧

3) $G(s) = \frac{K}{s^2(T_1s+1)(T_2s+1)}$

$G(s)$ 正实部极点数 $P=0$

$N_+ = 0 \quad N_- = 0 \quad R = 2(N_+ - N_-) = 0$

$Z = P - R = 0$ 系统闭环稳定, 有 2 个正实部极点

$V=2$, 从奈氏曲线上 $W=0^+$ 的对应点起逆时针补作 180° 且半径为无穷大的虚圆弧

4) $G(s) = \frac{K(T_1s+1)}{s^2(T_2s+1)}$

$G(s)$ 正实部极点数 $P=0$

$V=2$, 从奈氏曲线上 $W=0^+$ 的对应点起逆时针补作 180° 且半径为无穷大的虚圆弧

$N_+ = 0 \quad N_- = 0 \quad R = 0 \quad Z = P - R = 0$ 系统闭环稳定

5) $G(s) = \frac{K}{s^3}$

$G(s)$ 正实部极点数 $P=0$

$V=3$, 从奈氏曲线上 $W=0^+$ 的对应点起逆时针补作 270° 且半径为无穷大的虚圆弧

$N_+ = 0 \quad N_- = 0 \quad R = -2$

$Z = 2$ 系统闭环不稳定, 有 2 个正实部极点

6) $G(s) = \frac{K(T_1s+1)(T_2s+1)}{s^3}$

$G(s)$ 正实部极点数 $P=0$

$V=3$, 从奈氏曲线 $W=0^+$ 的对应点起逆时针补作 270° 且半径为无穷大的虚圆弧

$N_+ = 1 \quad N_- = 1 \quad R = 0 \quad Z = 0$ 系统闭环稳定

7) $G(s) = \frac{K(T_1s+1)(T_2s+1)}{s(T_1s+1)(T_2s+1)(T_3s+1)(T_4s+1)}$

$G(s)$ 正实部极点数 $P=0$

$V=1$, 从 $W=0^+$ 的对应点起逆时针补作 90° 的半径为无穷大的虚圆弧

$N_+ = 1 \quad N_- = 1 \quad R = 0 \quad Z = 0$ 系统闭环稳定

$$(8) \quad G(s) = \frac{K}{Ts-1}$$

$$P=1$$

$$N_+ = 0 \quad N_- = \frac{1}{2} \quad R = 2(N_+ - N_-) = -1$$

$$Z = P - R = 2 \quad \text{系统闭环稳定}$$

$$(9) \quad G(s) = \frac{-K}{-Ts+1}$$

$$P=1$$

$$N_+ = 0 \quad N_- = 0 \quad R = 0$$

$$Z = P - R = 1 \quad \text{系统闭环不稳定, 有一个正实部极点}$$

$$(10) \quad G(s) = \frac{K}{s(Ts-1)}$$

$$P=2$$

$N_+ = 0 \quad N_- = 1$, 从 $w=0$ 处对应点逆时针补作 90° 且半径为无穷大的虚圆弧

$$N_+ = 0 \quad N_- = \frac{1}{2} \quad R = -1$$

$$Z = P - R = 3 \quad \text{系统闭环不稳定, 有 2 个正实部极点}$$

5-15

$$\text{解: } G(s)H(s) = \frac{10}{s(s+1)(s^2+4)} = \frac{40}{s(s^2+4)(s+1)}$$

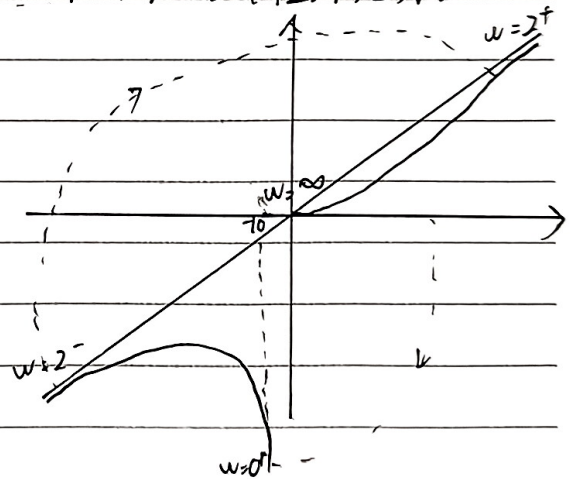
$$P=0$$

$N=1$, $N_+ = 1$, 从 $w=0^+$ 处对应点逆时针补作 90° 且半径为无穷大的虚圆弧; 从 $w=2j$ 处对应点起顺时针补作 180° 且半径为无穷大的虚圆弧

$$N_+ = 0 \quad N_- = 1 \quad R = 2(N_+ - N_-) = -2$$

$$Z = P - R = 2$$

系统闭环不稳定, 有 2 个正实部极点



5-16

解: $G(s) = \frac{K}{s(Ts+1)(s+1)}$

~~极点~~ $\rightarrow$ 令 $s = j\omega$ $G(j\omega) = \frac{K}{j\omega(jT\omega+1)(j\omega+1)}$
 $= -\frac{K(T+1)}{(1+T^2\omega^2)(1+\omega^2)} - j\frac{K(1-T\omega^2)}{\omega(1+T^2\omega^2)(1+\omega^2)}$

起点: $\omega=0^+$ 时 $G(j\omega) = -K(T+1) - j\infty$

终点: $\omega=\infty$ 时 $G(j\omega) = 0$

相角变化: $\angle(j\omega) : -90^\circ \sim -90^\circ$
 $\angle(\frac{1}{Ts+1}) : 0^\circ \sim -90^\circ$
 $\angle(\frac{1}{s+1}) : 0^\circ \sim -90^\circ$
 $\angle(K) = 0^\circ \sim 0^\circ$
 $\varphi(\omega) : -90^\circ \sim -270^\circ$

与实轴交点: $\text{Im}[G(j\omega_x)] = 0$ 即 $\frac{K(1-T\omega_x^2)}{\omega_x(1+T^2\omega_x^2)(1+\omega_x^2)} = 0$

$\omega_x = \sqrt{\frac{1}{T}}$
 $G(j\omega_x) = -\frac{KT}{1+T}$

绘制概略开环幅相特性曲线:

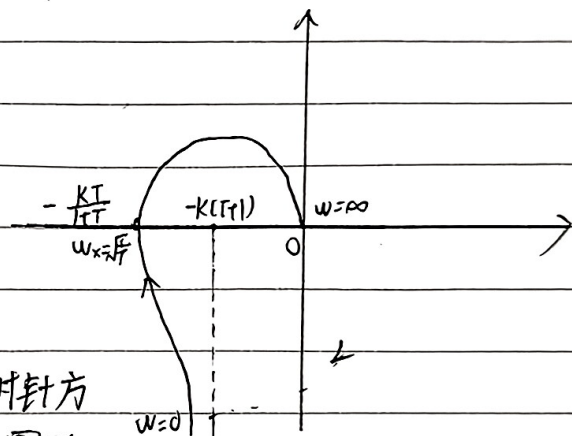
$\therefore T, K > 0$

$\therefore$ 每 $G(s)$ 正实部极点数 $P=0$

1) ~~$T=2$ 时, 若系统闭环稳定~~

$\nu=1$, 从奈氏曲线 $\omega=0^+$ 处对应点逆时针方向旋转补作 90° 且半径为无穷大的虚圆弧

2) $T=2$ 时, 若系统闭环稳定, 则 $\frac{KT}{1+T}$ 应位于 (-1, 0) 侧, 即 $-\frac{KT}{1+T} > -1$



代入 $T=2$, 得 $K < 1.5$, 则 K 的范围为 $0 < K < 1.5$

3) $K=10$ 时, 若系统闭环稳定, 则有 $-\frac{KT}{1+T} > -1$ 代入 $K=10$, 得 $T < \frac{1}{9}$
 则 T 的范围为 $0 < T < \frac{1}{9}$

3) 若系统闭环稳定, 则 $-\frac{KT}{1+T} > -1$

$0 < K < 1 + \frac{1}{T}$

$0 < T < \frac{1}{K-1}$

5-20

解: $G(s) = \frac{5s^2 e^{-\tau s}}{(s+1)^4}$

$$s = j\omega \quad G(j\omega) = \frac{-5\omega^2}{(1+j\omega)^4} e^{-\tau j\omega} = \frac{5\omega^2}{\sqrt{1+\omega^2}^4} (e^{j\arctan\omega})^4 e^{-\tau j\omega}$$

$$= \frac{-5\omega^2}{(1+\omega^2)^2} \frac{(1-j\omega)^2}{(1+j\omega)^2} e^{-\tau j\omega} = \frac{5\omega^2}{(1+\omega^2)^2} e^{-j(-\pi + 2\omega\tau + 4\arctan\omega)}$$

$$= \frac{-5\omega^2}{(1+\omega^2)^2}$$

起点: $\omega=0$ 时 $G(j0) = -5$

终点: $\omega=\infty$ 时 $G(j\infty) = 0$

与实轴交点: $-(-\pi + 2\omega\tau + 4\arctan\omega) = k\pi$

开环系统幅频特性 $|G(j\omega)| = \frac{5\omega^2}{(1+\omega^2)^2}$

求其极值可得: $\omega=1\text{rad/s}$ 时 $|G(j\omega)|_{\max} = 1.25$, 当 $\omega>1$ 时, 随着 ω 增大 $|G(j\omega)|$ 越来越小.

开环系统相频特性

$$\angle G(j\omega) = \pi - \tau\omega - 4\arctan\omega$$

随着 ω 增大, $\angle G(j\omega)$ 不断减小.

为保证闭环系统的稳定性, $\angle G(j\omega) = -180^\circ$, $|G(j\omega)|_{\max}$ 所对应的点频率 $\omega_{\max}$ 应当大于 1rad/s

(闭环稳定的一个条件: $\angle G(j\omega) = -\pi$ 时 $|G(j\omega)| < 1$)

(若 $\omega_{\max} < 1\text{rad/s}$, 可能在 $\angle G(j\omega)$ 未到 -180° 时, 幅频特性会使奈氏曲线包围 $(-1, j0)$ 点)

令 $\omega = \omega_m$ 时, 曲线与负实轴相交, 交点为 $(-\frac{5\omega_m^2}{(1+\omega_m^2)^2}, j0)$, 当 $\omega>1$ 时随着 ω 增大, 曲线与负实轴交点越来越接近原点

$G(s)$ 为正实部极点数 $P=0$, 若使 $Z=P-R=0$, 则 $R=2N=0$, 即 $N=0$

令 $-\frac{5\omega_m^2}{(1+\omega_m^2)^2} = -1$, 解得 $\omega_m = 1.618$ 或 0.618 (舍)

将 $\omega_m = 1.618$ 代入 $\angle(G(j\omega)) = -\pi$

$$\tau = 1.369$$

则 $0 < \tau < 1.369$ 时 系统闭环稳定

5-22

解: 典型二阶系统的开环传递函数为: $G(s) = \frac{K}{s(Ts+1)}$

$$\omega_n = \sqrt{\frac{K}{T}} \quad \zeta = \frac{1}{2\sqrt{TK}}$$

$$\because \omega_n = 3 \quad \zeta = 0.7$$

$$\therefore K = \frac{3}{\sqrt{1.4}} \quad T = \frac{1}{4.2}$$

$$\text{即 } G(s) = \frac{9}{s(s+4.2)}$$

$$\text{令 } s=j\omega, G(j\omega) = \frac{9}{j\omega(j\omega+4.2)} = \frac{9(-\omega+4.2j)}{\omega^3+4.2^2\omega} = \frac{9}{\omega\sqrt{17.64+\omega^2}} e^{-j(\frac{\pi}{2}+\arctan\frac{\omega}{4.2})}$$

$$|G(j\omega_c)| = 1, \text{ 即 } \frac{9}{\omega_c\sqrt{17.64+\omega_c^2}} = 1$$

$$\omega_c = 1.94 \text{ rad/s}$$

则截止频率为 1.94 rad/s

$$\text{相角裕度 } \gamma = \pi + \varphi(\omega_c)$$

$$= \pi + (-\frac{\pi}{2} - \arctan\frac{\omega_c}{4.2})$$

$$= 65.21^\circ$$